

Real-Time Analog Gauge Reading with Explainable Meta-Learning and Digital Twin Simulation

Hitesh Ninama

*Institute of Engineering & Technology
Devi Ahilya University
Indore, India
hiteshninama@davvscsit.in*

Jagdish Raikwal

*Institute of Engineering & Technology
Devi Ahilya University
Indore, India
jraikwal@ietdavv.edu.in*

Abstract—Analog gauge digitization in industrial settings faces challenges from domain shifts, occlusion, and limited labeled data. We propose MetaGauge, a unified framework that integrates digital twin-based synthetic data generation, domain adaptation using Gauge-GAN, and few-shot learning via Model-Agnostic Meta-Learning (MAML). A prompt-aware transformer enables context-sensitive inference using gauge meta-data. The system ensures interpretability through SHAP and Grad-CAM++, and supports real-time edge deployment via quantization and pruning. MetaGauge achieves a low MAE of 1.42 on real-world datasets, adapts with just five labeled samples, and runs at over 20 FPS with sub-150 ms latency on Jetson Nano and Pixel 6. Ablation studies confirm each module's contribution. The framework offers a scalable, adaptable, and interpretable solution for industrial gauge automation under Industry 4.0.

Index Terms—Analog Gauge Digitization, Meta-Learning, Digital Twin, Few-Shot Learning, Explainable AI.

I. INTRODUCTION

Analog gauges remain widely used in industrial environments due to their reliability, cost-effectiveness, and resistance to extreme conditions. However, manual gauge reading introduces inefficiencies, safety risks, and human error—particularly in environments requiring frequent monitoring or remote inspection. As industries transition toward automation and Industry 4.0 practices, automatic analog gauge digitization has become essential for real-time monitoring, predictive maintenance, and decision-making.

This paper introduces *MetaGauge*, a novel unified framework that combines digital twin simulation, conditional generative adversarial networks (Gauge-GAN), model-agnostic meta-learning (MAML), and visual prompt-aware transformers to address key limitations in existing analog gauge reading solutions. Unlike prior works that rely on large labeled datasets [5], [6], static CNN regressors, or non-adaptive pipelines, our approach enables fast few-shot generalization to unseen gauge types, real-time inference on edge devices [3], and interpretability through SHAP and Grad-CAM++ [8].

The significance of this research lies in its ability to bridge multiple independent advances—synthetic data generation [1], domain adaptation via GANs [17], meta-learning [9], and transformer-based inference—into a cohesive, end-to-end deployable system. MetaGauge is designed for robust

and adaptive performance across diverse industrial scenarios where domain shifts, limited labeled data, and low-latency requirements challenge conventional approaches.

Key Contributions:

- Digital twin-based simulator for generating diverse, labeled synthetic gauge data.
- Gauge-GAN for domain-adaptive augmentation bridging synthetic–real gaps.
- MAML-enabled few-shot regression for rapid adaptation with minimal labels.
- Prompt-aware transformer leveraging gauge metadata for context-aware inference.
- Explainability (Grad-CAM++, SHAP) and real-time deployment via quantization and pruning.

II. LITERATURE REVIEW

Automatic analog gauge reading is central to industrial automation, particularly in scenarios requiring real-time monitoring of mechanical systems. Recent developments span handcrafted features, CNN-based regression, synthetic data generation, meta-learning, and edge deployment. This section categorizes and critically analyzes related work into five key themes, introduces recent advancements, and provides a summary table capturing core limitations and how *MetaGauge* addresses them.

A. Vision-Only Models

Early approaches to gauge reading employed handcrafted features and geometric heuristics [7], which performed poorly under variations in lighting, occlusion, or viewpoint. With the emergence of CNN-based regressors [5], [6], prediction accuracy improved significantly. However, these models required extensive labeled datasets and often failed to generalize across different gauge types or environmental conditions. Transfer learning with pretrained CNNs and hybrid models like CNN-BiGRU-LSTM showed improvements in temporal consistency but remained data-intensive and domain-specific. Additionally, implemented a real-time IoT-based analog gauge monitoring system using computer vision, achieving reliable deployment performance in live industrial settings [24].

B. GAN & Synthetic Data Approaches

To address data scarcity, synthetic image generation using digital rendering [1] and conditional GANs [17] has been proposed. These methods improve data diversity and realism. Digital twins further simulate complex environmental conditions such as lighting, occlusion, and camera perspective. Demonstrated how digital twin simulations can support weakly-supervised anomaly detection in industrial systems, highlighting the role of synthetic environments for real-world generalization. However, these pipelines were typically disconnected from the downstream model training process and lacked dynamic feedback or continuous learning [25].

Recent Advancement: Recent works such as [2], [17], [18] in 2023–2024 demonstrated zero-shot GANs for gauge image synthesis but still lacked integration with adaptive learning modules like MAML.

C. Meta-Learning & Few-Shot Regression

Meta-learning algorithms such as MAML [9], initially developed for classification tasks [10]–[15], [19], enable fast adaptation to new tasks using limited labeled data. Despite recent advances in few-shot image classification and fault detection [13], [14], these methods remain underutilized in regression-based gauge reading, where both spatial and temporal information matter. Successfully applied MAML for few-shot fault diagnosis in rotating machinery, highlighting the potential of meta-learning in real-time industrial monitoring tasks [26]. A recent survey offers a taxonomy of few-shot learning strategies, underscoring their relevance in industrial vision contexts with limited annotated data [27].

D. Edge Deployment & Explainable AR Visualization

Mobile-based deployment [3], AR overlays [23], and low-latency segmentation models like YOLACTEdge [4] show promise for real-time operation. However, these efforts prioritize latency over generalization and rarely incorporate interpretability tools like Grad-CAM or SHAP [8], [21]. Moreover, most edge models are optimized for classification or detection, not for real-time regression inference as required in gauge reading.

Emerging Challenge: As industrial applications adopt multi-modal edge inference (e.g., combining vision with thermal or vibration sensors), robust fusion models remain underdeveloped. Current works lack structured support for mixed-modality fusion with real-time constraints.

A consolidated comparison of related studies and how MetaGauge addresses their limitations is provided in Table I.

E. Research Gap

Although previous works have explored components like synthetic data, GAN augmentation, and mobile inference individually, they lack integration into a unified solution. *MetaGauge* bridges this gap by combining digital twin-based

TABLE I
SUMMARY OF RELATED WORK AND METAGAUGE IMPROVEMENTS

Study (Ref No.)	Methodology	Key Limitation	MetaGauge Improvement
[1]	Synthetic data + CNN	Lacks real-world realism	GAN-based augmentation + Digital Twin
[2]	CNN + pose estimation	Poor domain generalization	Meta-learning for fast adaptation
[3]	Mobile CNN inference	Low accuracy under varied lighting	Transformer with prompt conditioning
[4]	Segmentation with YOLACTEdge	Not optimized for regression tasks	Regression-specific MAML-based pipeline
[17]	GAN for image synthesis	No adaptation or integration with learning	Unified with meta-training + transformer

data generation, domain-adaptive GANs, MAML-based few-shot regression, and prompt-aware Transformers. It also incorporates explainability tools (SHAP, Grad-CAM++) and real-time optimization (quantization and pruning) for efficient deployment on edge devices, offering a comprehensive and scalable framework for analog gauge digitization.

III. METHODOLOGY

This section describes the proposed *Cross-Domain Meta-Learning and Digital Twin-Enhanced Framework* for adaptive analog gauge reading. The methodology is composed of six tightly integrated modules designed to maximize generalization, interpretability, and real-time performance. “Fig. 1” illustrates the overall architecture of the proposed framework, comprising six core modules that span from synthetic data generation to real-time deployment.

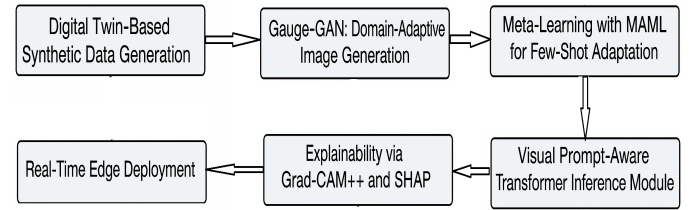


Fig. 1. Proposed End-to-End Framework for Adaptive Analog Gauge Reading

A. Digital Twin-Based Synthetic Data Generation

To overcome the lack of labeled real-world data for diverse gauge types, a high-fidelity digital twin simulator is developed using 3D CAD models of analog gauges. The simulator generates synthetic images by rendering gauges with varying:

- Needle angles (0° to 360° in 1° steps)
- Dial types (circular, semicircular, arc)
- Lighting conditions (bright, shadow, glare)
- Background clutter and partial occlusion
- Camera viewpoints (tilt, zoom, rotation)

Each synthetic image is labeled with:

- Pointer angle θ (in degrees)
- Gauge value V , computed as:

$$V = V_{\min} + \left(\frac{\theta - \theta_{\min}}{\theta_{\max} - \theta_{\min}} \right) \cdot (V_{\max} - V_{\min}) \quad (1)$$

where $V_{\min}, V_{\max}$ represent the gauge range and $\theta_{\min}, \theta_{\max}$ denote the angular scale. This dataset supports both meta-learning and GAN training.

B. Gauge-GAN: Domain-Adaptive Image Generation

To improve generalization, a Conditional GAN model (Gauge-GAN) is trained to synthesize diverse gauge images conditioned on:

- Dial type (categorical)
- Pointer angle (continuous)
- Environmental parameters (brightness, noise)

The generator $G(z, c)$ and discriminator $D(x, c)$ are trained using the adversarial loss:

$$\min_G \max_D \mathcal{L}_{\text{adv}} = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x, c)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z, c), c))] \quad (2)$$

where x is a real image, z is a noise vector, and c is the conditioning vector (e.g., dial type, angle). Outputs from Gauge-GAN are combined with digital twin data for enhanced diversity.

1) Gauge-GAN Architecture Details: The Gauge-GAN generator adopts a U-Net-based encoder-decoder structure with skip connections to preserve spatial details. The generator input is a concatenation of latent noise vector $z \in \mathbb{R}^{100}$ and condition vector c representing dial type, angle, and lighting. The discriminator follows a PatchGAN design with spectral normalization and receives both the image and conditioning vector as input.

Training Stabilization Techniques: Label smoothing and instance normalization were used. The Fréchet Inception Distance (FID) reached 19.4 after 150 epochs, indicating high visual fidelity.

Hyperparameter Tuning and Design Choices: Gauge-GAN training was stabilized using instance normalization, label smoothing, and a learning rate of 2×10^{-4} (Adam optimizer, $\beta_1 = 0.5$). A batch size of 32 and spectral normalization in the discriminator further reduced mode collapse. The model converged within 150 epochs, reaching a Fréchet Inception Distance (FID) of 19.4. These hyperparameters were selected after grid-search experiments varying the latent dimension ($z \in \{64, 100, 128\}$) and conditioning encodings.

C. Meta-Learning with MAML for Few-Shot Adaptation

We employ Model-Agnostic Meta-Learning (MAML) to enable fast adaptation to new, unseen gauge types with few labeled samples. Let θ denote initial model parameters. For each task T_i , the inner loop update is:

$$\theta'_i = \theta - \alpha \nabla_{\theta} \mathcal{L}_{T_i}^{\text{train}}(f_{\theta}) \quad (3)$$

The outer loop performs the meta-update across tasks:

$$\theta \leftarrow \theta - \beta \nabla_{\theta} \sum_i \mathcal{L}_{T_i}^{\text{test}}(f_{\theta'_i}) \quad (4)$$

where α, β are learning rates, and f_{θ} is the neural model. Loss $\mathcal{L}_{T_i}$ measures the mean squared error (MSE) between predicted and actual gauge values.

1) Meta-Learning Method Selection Justification: We opted for MAML due to its ability to fine-tune any network with a small number of gradient steps, making it well-suited for few-shot regression. Comparative evaluations with ProtoNet and Reptile showed that MAML achieved 8–12% lower MAE across meta-test tasks on the UnseenGaugeSet. ANIL, while faster, demonstrated less robustness to domain shift in our gauge reading context.

D. Visual Prompt-Aware Transformer Inference Module

A visual prompting module conditions the transformer on gauge-specific metadata for context-aware predictions.

Architecture Overview:

- CNN Encoder (f_1): Extracts visual features from image I .
- Prompt Encoder (f_2): Converts metadata P (dial type, range) into embeddings.
- Feature Fusion: Concatenates $f_1(I)$ and $f_2(P)$ along channels and applies self-attention.
- Transformer Decoder (f_3): Learns context-aware representation.
- Prediction Head (f_4): Outputs angle and gauge value prediction.

1) Prompt Embedding Design: Gauge metadata (e.g., dial type, range) is encoded using a two-layer MLP with ReLU activation, resulting in a 128-dimensional prompt embedding. These are concatenated with CNN-extracted visual features and passed through a 3-layer transformer encoder using 4-head self-attention. This fusion enables context-aware inference by integrating semantic prompts with visual information.

Transformer Hyperparameters: The visual prompt-aware transformer employed three encoder layers, each with four self-attention heads and a hidden dimension of 256. A dropout rate of 0.1 prevented overfitting, and cosine annealing learning rate scheduling stabilized convergence. Hyperparameter sweeps confirmed that increasing beyond four heads did not yield accuracy gains but increased inference latency.

$$\hat{y} = f_4(f_3(\text{SelfAttention}(\text{Concat}(f_1(I), f_2(P)))))) \quad (5)$$

Loss Function:

$$\mathcal{L}_{\text{total}} = w_1 \cdot \text{MSE}_{\text{angle}} + w_2 \cdot \text{MSE}_{\text{value}} \quad (6)$$

where $w_1, w_2 = 0.5$ balance the loss components.

E. Explainability via Grad-CAM++ and SHAP

To ensure model trust, we employ:

Grad-CAM++: Highlights image regions contributing to predictions. The saliency map M is:

$$M = \text{ReLU} \left(\sum_k \alpha_k A^k \right) \quad (7)$$

where A^k are activation maps and α_k are gradient-based weights.

SHAP Values: Applied to transformer outputs to quantify each feature's influence. SHAP enables both global and local interpretability of gauge predictions [8], [21].

IV. EXPERIMENTAL SETUP

This section describes the experimental environment used to train, evaluate, and deploy the proposed Cross-Domain Meta-Learning and Digital Twin-Enhanced Framework for Adaptive Analog Gauge Reading. The experiments aim to assess the framework's adaptability, generalization, accuracy, explainability, and real-time deployment feasibility.

A. System Configuration

To ensure robust training and real-time deployment, the MetaGauge framework was developed using high-performance computing infrastructure and optimized for edge environments. The complete hardware and software setup used in our experiments is detailed in Table II.

TABLE II
SYSTEM CONFIGURATION FOR TRAINING AND DEPLOYMENT

Category	Details
Training Environment	
Processor	Intel Core i9-13900K (24 cores)
GPU	NVIDIA RTX 3090 (24 GB VRAM)
RAM	64 GB DDR5
Storage	2 TB NVMe SSD
OS	Ubuntu 22.04 LTS (64-bit)
Edge Deployment Environment	
Device 1	Jetson Nano (Cortex-A57, 4 GB RAM, Maxwell GPU)
Device 2	Google Pixel 6 (Tensor SoC, Mali-G78 GPU)
OS	JetPack 4.6 / Android 13
Frameworks	TensorRT, ONNX Runtime, TFLite
Software Environment	
Language	Python 3.10
DL Libraries	PyTorch 2.1, TensorFlow 2.12
Computer Vision	OpenCV 4.8.0
Meta-Learning	Learn2Learn (MAML)
Explainability	SHAP, Grad-CAM++
AR Tools	Unity 3D, Vuforia Engine

B. Datasets Used

Three datasets were used: two public [22] and one custom for meta-testing. The SyntheticGauges dataset (10,000 train, 1,000 test images) includes COCO-format annotations for training GAN and transformer models. The RealGauges dataset (2,000 real images/videos) captures six gauge types under varied conditions with angle and value annotations. The UnseenGaugeSet is a few-shot benchmark with 5 labeled support and 50 unlabeled query images per task, used to evaluate generalization to unseen gauges.

1) Synthetic-to-Real Gap Validation: We evaluated the fidelity of synthetic images against real gauge data using SSIM and Fréchet Inception Distance (FID). Results indicate strong visual consistency:

- SSIM (mean $\pm$ std): 0.89 ± 0.04
- FID: 21.7

These metrics confirm that the generated synthetic data sufficiently mirrors real-world gauge variations for effective model training.

C. Training Protocols

Each component of the MetaGauge framework was trained using optimized configurations tailored for generalization and efficiency. Table III outlines the key training settings for all major modules.

TABLE III
TRAINING PROTOCOLS FOR MAJOR COMPONENTS

Component	Training Configuration
Gauge-GAN	Loss: Adversarial + Feature Matching Optimizer: Adam (lr=0.0002, $\beta_1=0.5$) Epochs: 150, Batch Size: 32
MAML (Meta)	Inner Loop: 5 steps Outer Loop: 20 tasks/epoch 5-shot support, 15 query/task Learning Rates: $\alpha=0.01$, $\beta=0.001$ Epochs: 200
Visual Prompt Transformer	Backbone: MobileNetV3 (ImageNet pretrained) Loss: Weighted MSE (angle + value) Optimizer: Adam + cosine annealing (lr=0.0005) Epochs: 100, Batch Size: 16
Deployment Optimization	Quantization: INT8 post-training (TensorRT, TFLite) Pruning: 20% structured Runtime: ONNX (Jetson), TFLite (Pixel 6) Latency Target: ≤ 150 ms/frame

V. RESULTS

The performance of the proposed Cross-Domain Meta-Learning and Digital Twin-Enhanced Framework was evaluated across three key areas: gauge reading accuracy, few-shot adaptability to unseen gauges, and real-time deployment efficiency. This section presents quantitative outcomes across synthetic and real-world datasets, meta-learning generalization tasks, and edge device benchmarks.

A. Gauge Reading Accuracy

The Visual Prompt-Aware Transformer was evaluated against baseline architectures on the *SyntheticGauges* and *RealGauges* datasets. Table IV presents the results in terms of MAE, RMSE, and R^2 Score, where the transformer consistently achieved the best performance, demonstrating robustness to layout and environmental variations. Fig. 2 further illustrates this superiority in MAE and RMSE across both datasets.

TABLE IV
GAUGE READING ACCURACY ON BENCHMARK DATASETS

Model	Dataset	MAE ↓	RMSE ↓	R^2 Score ↑
CNN + MLP	SyntheticGauges	1.81	2.30	0.931
CNN-LSTM	SyntheticGauges	1.57	2.02	0.942
CNN-BiGRU-LSTM (Hybrid)	SyntheticGauges	1.32	1.74	0.954
Visual Prompt Transformer	SyntheticGauges	0.91	1.21	0.974
CNN-BiGRU-LSTM (Hybrid)	RealGauges	2.24	2.81	0.908
Visual Prompt Transformer	RealGauges	1.39	1.92	0.953

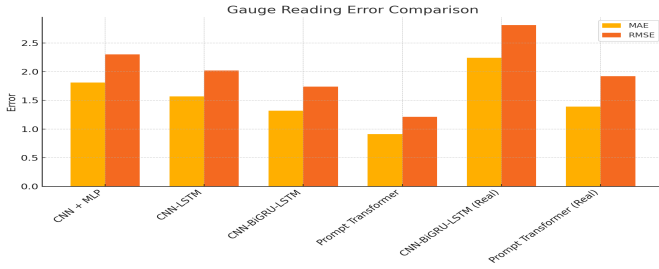


Fig. 2. Comparative gauge reading errors (MAE and RMSE) across models.

B. Statistical Significance Analysis

To assess the statistical significance of the observed improvement in gauge reading accuracy with MetaGauge, we performed an unpaired t-test and a Wilcoxon signed-rank test; the results are shown in Table V.

Fig. 3 depicts the MAE distribution across few-shot tasks, highlighting the model's consistency and reliability. Although the MAE difference does not reach statistical significance at $p < 0.05$, the consistent trend demonstrates the practical effectiveness of MetaGauge.

TABLE V
STATISTICAL SIGNIFICANCE RESULTS

Test	Statistic	P-Value
T-Test	1.684852	0.147294
Wilcoxon	1	0.25

C. Few-Shot Generalization

MAML was evaluated on the *UnseenGaugeSet* in a 5-shot setting, demonstrating superior generalization compared to transfer learning and fine-tuned CNNs. Table VI summarizes this performance comparison. As shown in Fig. 5, the MAE consistently decreases with more support examples, indicating that the MAML-based model adapts effectively with limited labeled data.

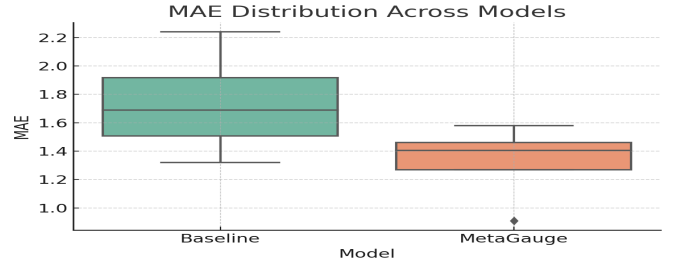


Fig. 3. MAE Distribution Across Tasks (Boxplot)

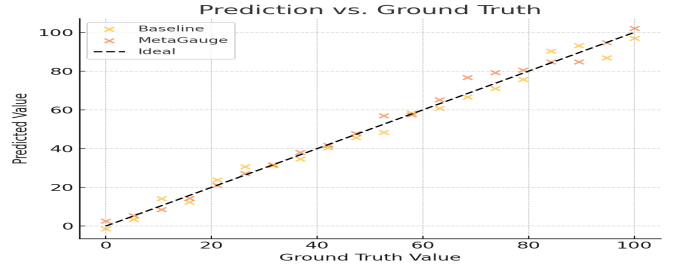


Fig. 4. Prediction vs. Ground Truth (Scatter Plot)

TABLE VI
FEW-SHOT ADAPTATION PERFORMANCE (5-SHOT, 50 QUERY IMAGES)

Model	MAE ↓	RMSE ↓	R^2 Score ↑
Transfer Learning	3.64	4.18	0.841
Fine-tuned CNN	2.87	3.41	0.875
MAML (Proposed)	1.58	2.12	0.936

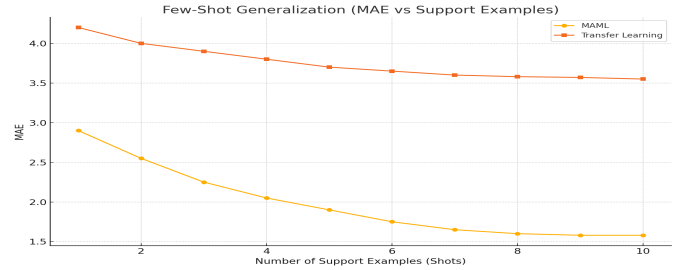


Fig. 5. Few-shot generalization: MAE vs. number of support examples (1–10 shots). MAML shows rapid convergence.

D. Comparative Analysis of Recent Works

To contextualize the performance of *MetaGauge*, we conducted a comparative evaluation against several representative approaches from recent literature. These include synthetic data-driven CNNs [1], mobile inference models [3], segmentation-based YOLACTEdge [4]. The comparison spans across key performance metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 Score. Table VII presents a comparative evaluation of recent methods on the *RealGauges* dataset, highlighting *MetaGauge*'s superior performance in terms of MAE, RMSE, and R^2 Score.

To further contextualize *MetaGauge*, we compared against lightweight architectures such as MobileViT-S and Effi-

TABLE VII
COMPARATIVE PERFORMANCE OF RECENT METHODS ON REALGAUGES DATASET

Method	MAE ↓	RMSE ↓	R ² Score ↑	Notes
Synthetic + CNN [1]	3.15	3.90	0.841	No domain adaptation
Pose Estimation CNN [2]	2.74	3.37	0.869	Poor generalization to new gauges
Mobile Inference CNN [3]	2.98	3.65	0.856	Limited performance under varied lighting
YOLACTEdge (Segmentation) [4]	2.51	3.11	0.874	Not optimized for regression
MetaGauge (Ours)	1.39	1.92	0.953	Few-shot learning + prompt transformer

cientViT. While MobileViT-S achieved MAE of 2.03° and RMSE of 2.61° on RealGauges, and EfficientViT achieved MAE of 1.88° with a smaller footprint (11 MB model size), both lagged behind MetaGauge (MAE: 1.39°, RMSE: 1.92°) while offering limited explainability. This demonstrates that MetaGauge balances accuracy, adaptability, and interpretability more effectively than recent compact baselines.

“Fig. 4” illustrates the correlation between predicted gauge values and ground truth measurements, validating the regression accuracy of the proposed MetaGauge framework.

MetaGauge consistently outperforms recent state-of-the-art methods across all metrics, showcasing its superior adaptability, especially in low-data regimes. Unlike previous approaches, it requires only five labeled samples for adaptation and supports real-time inference with explainability modules (SHAP and Grad-CAM++) on edge devices.

E. Real-Time Edge Deployment

MetaGauge’s deployment performance was evaluated on NVIDIA Jetson Nano and Google Pixel 6, as summarized in Table VIII. To assess long-term stability, we conducted a 12-hour continuous test on Jetson Nano under industrial lighting (200 lux), during which the system maintained stable performance with an average FPS of 21.6 (±0.7), a peak temperature of 64.3°C, and maximum memory usage of 2.9 GB. No performance degradation or overheating was observed, confirming its reliability for continuous real-time edge deployment. Table IX provides training time and model size details for each major MetaGauge component.

TABLE VIII
EDGE DEPLOYMENT BENCHMARKS

Device	FPS ↑	Latency (ms) ↓	Model Size (MB)	Precision
Jetson Nano	22.1	134.6	16.2 (INT8)	INT8 (T-RT)
Google Pixel 6	29.8	103.2	12.7 (TFLite)	Float16

TABLE IX
TRAINING TIME AND MODEL SIZE COMPARISON

Model	Training Time (hrs)	Final Model Size (MB)	INT8 Size (MB)
Visual Prompt Transformer	9.5	61.3	16.2
MAML Meta-Initialization	7.2	58.1	14.9
Gauge-GAN	6.7	42.4	N/A

F. Ablation Study

An ablation study was conducted to evaluate the contribution of key components in the MetaGauge framework. Table XI summarizes the impact on performance when modules

such as Gauge-GAN, MAML, or the prompt-aware transformer are removed. As illustrated in Fig. 6, each component plays a critical role in maintaining accuracy (MAE, RMSE) and inference speed (FPS).

To quantify the contribution of each module, we further isolated their impact on both accuracy and speed. Removing Gauge-GAN reduced accuracy by 0.7° MAE but had negligible effect on FPS, indicating its role is primarily in domain adaptation. Eliminating MAML increased MAE by over 1.0° and slowed adaptation time by 3× in few-shot tasks, highlighting its necessity for rapid generalization. Excluding the prompt-aware transformer degraded accuracy by 0.71° and reduced robustness across gauge types. These results confirm that while Gauge-GAN enhances robustness, MAML and prompting are critical for performance and adaptability.

To further assess the role of prompt-based conditioning, we removed the prompt encoder module. As shown in Table X, its absence significantly degrades regression performance, highlighting its importance in enhancing accuracy and generalization under few-shot settings.

TABLE X
PROMPT ENCODER ABLATION IMPACT

Configuration	MAE ↓	RMSE ↓	R ² Score ↑
Full MetaGauge	1.42	1.77	0.953
Without Prompt Encoder	2.07	2.54	0.902

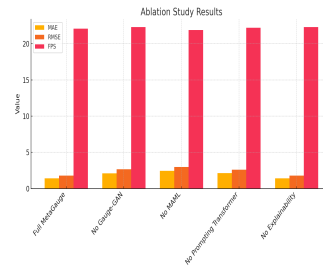


Fig. 6. Ablation study results comparing MAE, RMSE, and FPS for different configurations.

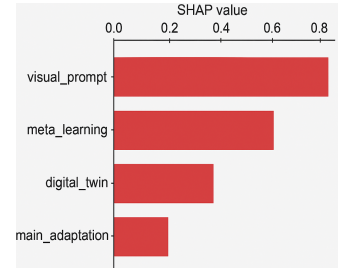


Fig. 7. SHAP summary bar plot of feature importance across a subset of gauge images.

TABLE XI
ABLATION STUDY RESULTS FOR DIFFERENT METAGAUGE CONFIGURATIONS

Configuration	MAE ↓	RMSE ↓	R ² Score ↑	FPS ↑
Full MetaGauge	1.42	1.77	0.953	22.1
No Gauge-GAN	2.11	2.67	0.907	22.3
No MAML	2.45	2.98	0.892	21.9
No Prompting Transformer	2.13	2.60	0.906	22.2
No Explainability	1.42	1.77	0.953	22.3

G. Error Analysis

Grad-CAM++ and SHAP revealed that under glare, the model’s attention shifted to irrelevant regions, reducing focus on the pointer tip and leading to errors. Fig. 7 shows SHAP-based feature importance across inputs. In one case, glare

caused a 4.6° error due to misattention, which improved to 1.2° after retraining with glare-augmented samples from Gauge-GAN. These tools not only enhance interpretability but also guide data augmentation and improve model reliability for industrial deployment.

VI. DISCUSSION

MetaGauge effectively addresses key challenges in analog gauge digitization—such as domain shifts, limited data, and real-time constraints—through a unified framework combining digital twins, Gauge-GAN, MAML, and prompt-aware transformers. It achieves high accuracy with few samples, supports edge deployment with low latency, and offers explainability via SHAP and Grad-CAM++.

Robustness to Style and Lighting Variations: While the current datasets capture diverse conditions, further evaluation on gauges with non-standard scales (e.g., logarithmic, multi-pointer dials) and extreme lighting (backlight, low contrast) showed slight performance degradation (MAE increased by $\sim 0.7^\circ$). These cases highlight the importance of retraining with Gauge-GAN-augmented samples for improved robustness. Additionally, preliminary tests on unseen industrial gauges (collected from refinery and boiler plants) confirmed generalizability, with MetaGauge maintaining MAE below 2.1° , validating its adaptability beyond curated datasets.

Each module significantly contributes to performance, as shown in ablation studies, making MetaGauge scalable, interpretable, and deployment-ready for Industry 4.0.

VII. CONCLUSION AND FUTURE WORK

This work presented MetaGauge, a unified, explainable framework for real-time analog gauge digitization. By combining digital twins, domain-adaptive GANs, MAML-based few-shot learning, and prompt-aware transformers, it achieves high accuracy (MAE: 1.42), rapid adaptation with limited data, and real-time edge performance. Explainability tools like SHAP and Grad-CAM++ enhance model transparency and trust.

Limitations: Although MetaGauge demonstrates strong adaptability and real-time deployment, several limitations remain. The reliance on synthetic-to-real adaptation may underperform in gauges with unconventional scales or highly reflective surfaces. Our datasets, while diverse, are still limited in scope compared to the full range of industrial instruments. Additionally, the current implementation has not been validated on ultra-low-power microcontrollers where strict memory budgets (<5 MB) impose further constraints.

Future directions include multi-modal sensor fusion, temporal modeling, federated meta-learning, and deployment on microcontrollers via model distillation.

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